

Bluetooth Audio Adaptor — Design Overview

Rev B (in design) · Updated 2026-08-30 · Jason Describes the **target design**. Deltas from as-built Rev A are listed in §10.

Living document. Every confirmed design change lands here. Planning, open questions and review findings live in 0 and 1 — not here.

1. Function

Bluetooth audio adaptor for a car AUX input. Powered from USB-C or an internal Li-ion cell.

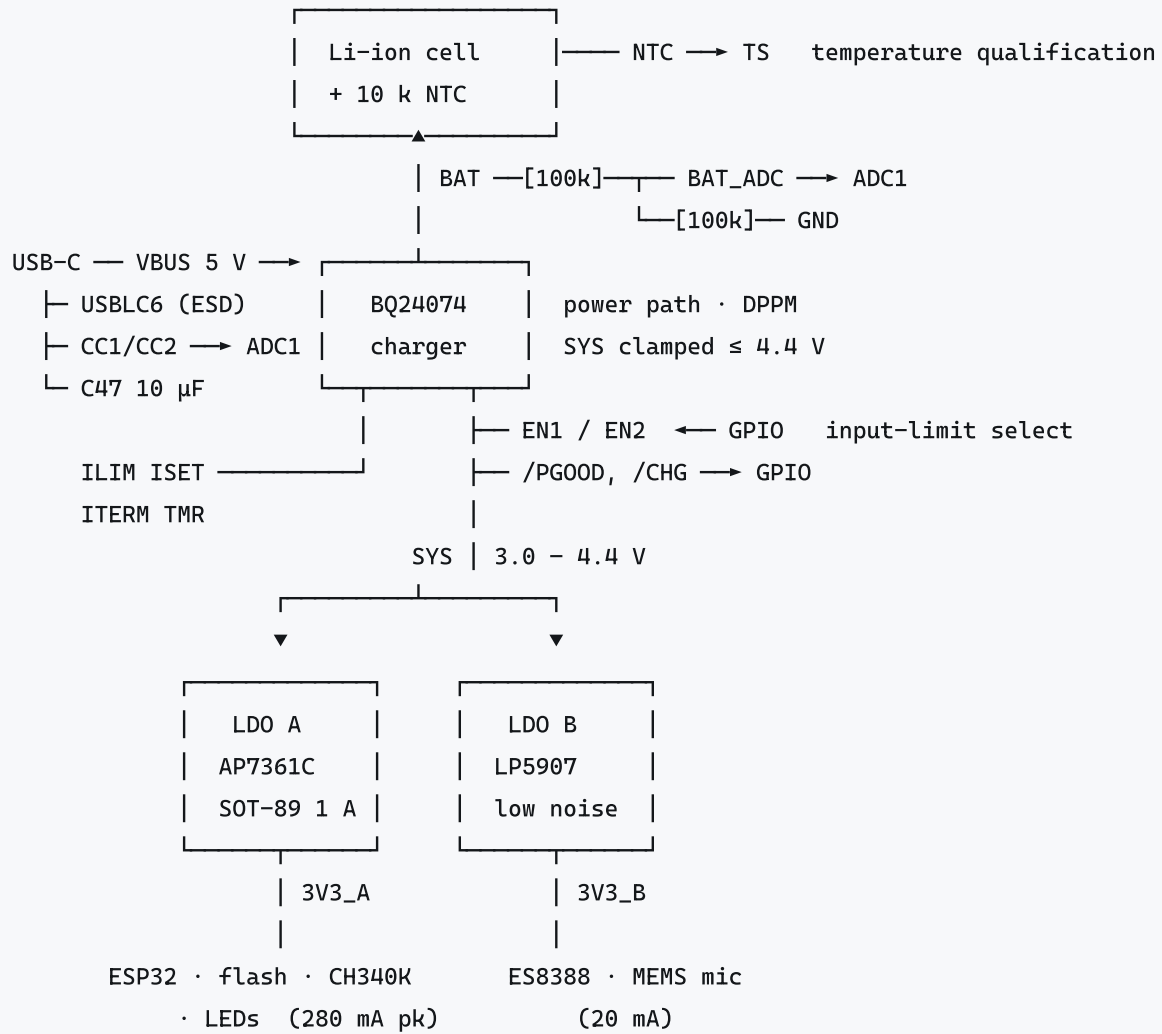
Mode	Path	Status
RX (primary)	Phone → Bluetooth A2DP → codec DAC → 3.5 mm out	Rev A hardware complete
TX	3.5 mm in → codec ADC → Bluetooth → headphones/speaker	Rev B addition
HFP handsfree	On-board mic → codec ADC → BT; BT → DAC → jack	Rev B addition

Key constraint: must work with all phones, not just recent ones → **Bluetooth Classic (A2DP/HFP), not BLE Audio**. This single requirement determines the MCU (\$5).

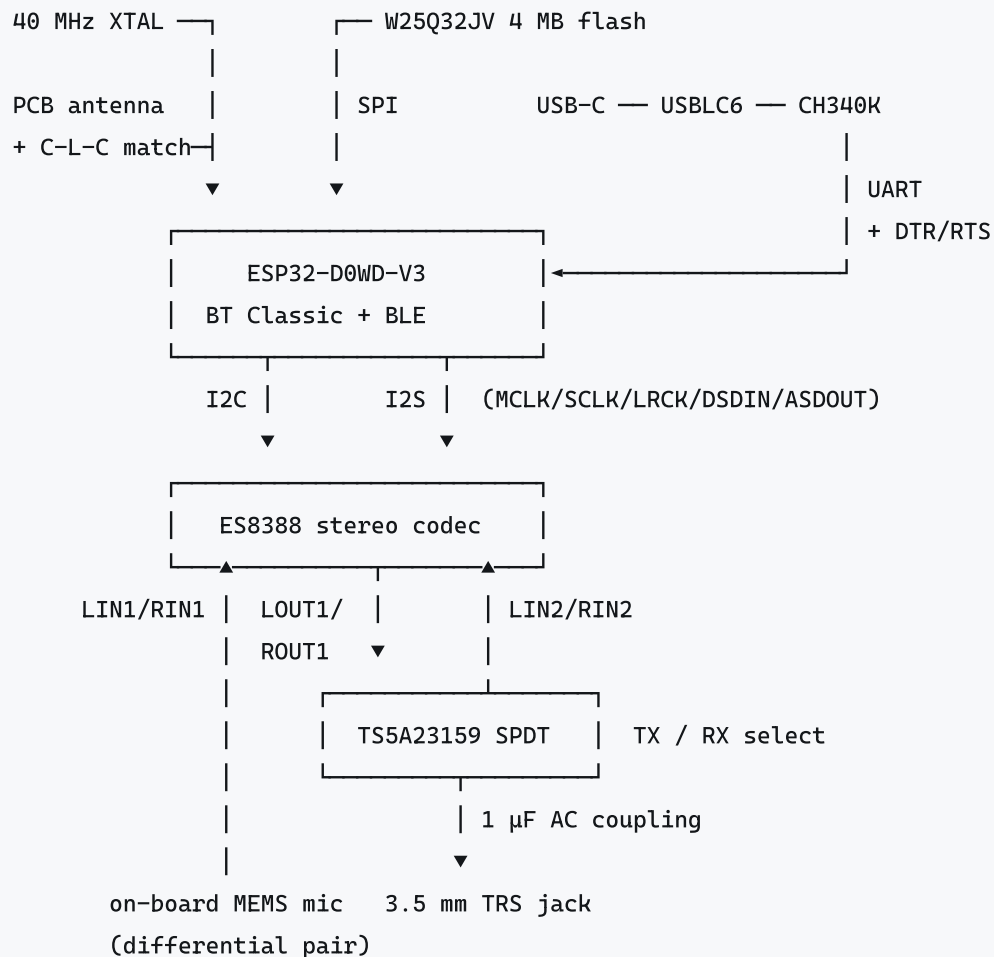
Out of scope: amplified speaker output, multipoint pairing, app control.

2. Architecture

Power tree



Signal chain



Rails

Rail	Source	Loads	Why separate
SYS 3.0–4.4 V	BQ24074	Both LDOs	Power path: USB when present, battery when not
3V3_A	LDO A	ESP32, flash, CH340K, LEDs	Carries RF current bursts
3V3_B	LDO B (LP5907)	ES8388, mic	Isolates a 96 dB codec from those bursts

3. Power subsystem

Linear regulation throughout. A switching converter was rejected: this is a mixed-signal audio board and the efficiency gain does not pay for the noise.

Part	Function	Key spec
BQ24074RGTR (C54313)	Power-path charger	1.5 A, SYS clamped ≤ 4.4 V, DPPM, NTC input, 10.5 V OVP
AP7361C-33Y5-13 (C460397)	LDO A — digital/RF	SOT-89-5, 1 A, ultra-low dropout
LP5907MFX-3.3 (C80670)	LDO B — codec analog	6.5 μV RMS, high PSRR, no bypass cap needed

Part	Function	Key spec
USBLC6-2SC6	USB ESD	In-line on D+/D-; its VBUS pin stays at the connector, upstream of the charger
Li-ion cell + 10 k NTC	Energy	Removable. NTC bonded to cell, not PCB

VBUS reaches exactly three things: the ESD array, the charger's **IN** pin, and **C47**. Every other load in the system hangs off **SYS**, including both LDOs. That is what makes the board behave identically on USB and on battery.

3.1 Power path — the five loops

In priority order. The system load always outranks the battery.

Loop	Trips at	Behaviour
Input current limit	$EN1 / EN2 + R_{ILIM}$	Hard ceiling on total current into IN . Everything else lives inside this budget
VIN-DPM	V_{IN} falls to 4.5 V	Reduces input current so a weak source cannot be crashed. Active only in USB100/USB500 modes
DPPM	V_{SYS} falls to $V_{O(REG)} - 100 \text{ mV} \approx 4.3 \text{ V}$	Cuts <i>charge</i> current to hold SYS up. Charge termination is disabled while active
Battery supplement	$V_{SYS} < V_{BAT} - 40 \text{ mV}$	Charge current already zero and the load still exceeds the input limit — BATFET turns fully on and the cell supplements. Exits above $V_{BAT} - 20 \text{ mV}$
Thermal regulation	$T_J \geq 125 \text{ }^{\circ}\text{C}$	Folds back charge current regardless of the above

The last three slow the safety timers **in proportion to the charge-current reduction**. A deliberately low **R_ISET** gets no such slowdown — the timer would run at full speed while the cell trickled. This is why **R_ISET** is set *above* anything the input can deliver, and DPPM is left to mediate (§3.3).

3.2 Input current limit — mode strapping

EN2 / EN1 select the limit. Datasheet Table 7-2 orders the columns EN2, EN1:

EN2	EN1	Limit	When
0	1	475 mA (USB500)	Power-on default — and any unknown or default-USB source
1	0	1.073 A (R_ILIM)	Firmware has confirmed a $\geq 1.5 \text{ A}$ source (§3.4)
0	0	95 mA (USB100)	Strict compliance on an un-enumerated host port
1	1	—	Standby / USB suspend

The default must be USB500, not USB100. Both pins floating gives USB100 through the chip's 285 k Ω internal pulldowns, and 100 mA cannot run a 171 mA system: with a flat cell the board would brown out before firmware ran and could never raise the limit. USB500 boots the system, matches what a PC port allows after enumeration, and is a mode that has VIN-DPM behind it.

SYS —[R30 100k]—┐ EN1 → GPIO (drive open-drain) default HIGH (3.26 V)
 |
 (285k internal pulldown)

GND —[R31 100k]—┐ EN2 → GPIO (push-pull) default LOW

100 kΩ, not 0 Ω — the GPIO must be able to override the resistor. Against the internal 285 kΩ, EN1 sits at $4.4 \times 285/385 = 3.26$ V, comfortably over the 1.4 V_{IH} .

Pull up to SYS, not 3V3_A — 3V3_A does not exist until the system has booted, and the whole point of the default is to be defined before that. SYS is present whenever either source is, and at 4.4 V max sits inside the EN pin's 1.4–6 V window (7 V absolute max).

Drive EN1 open-drain: the pull-up is to 4.4 V and the GPIO drives 3.3 V. Leave both pins alone whenever /PGOOD is high-Z — the EN inputs are ignored in power-down mode, and driving EN1 low would only burn cell current through R30.

3.3 Charge programming

Resistor	Value	Formula	Result
R_ISET	1.2 kΩ 1 %	$I_{CHG} = K_{ISET}/R$, $K = 890$	742 mA (664–812 over spread)
R_ILIM	1.5 kΩ	$I_{IN(max)} = K_{ILIM}/R$, $K = 1610$	1.073 A
R_TERM	3.0 kΩ	$I_{TERM} = 0.03 \times R_{ISET}$	75 mA ($\approx C/27$ on a 2000 mAh cell)
R_TMR	56 kΩ	$t_{MAXCHG} = 10 \times K_{TMR} \times R$, $K = 48 \text{ s/k}\Omega$	7.5 h (5.6 h worst case)

Pre-charge falls out of the same ISET resistor: $K_{PRECHG}/R_{ISET} = 88/1200 = \mathbf{73 \text{ mA}}$, the conventional 10 % of fast charge.

742 mA is 0.37C on a 2000 mAh cell and 0.25C on a 3000 mAh one — safe for either, so this resistor does not depend on the cell decision. Use 1 % on R_ISET: TI calls this out specifically to avoid tripping the internal Riset short-circuit test.

ILIM must never be left open — an open ILIM pin disables all charging. Worth an explicit bring-up check.

R_NTC_LIN is fitted **DNP** by design: it stands in for the pack NTC if a cell without one is used, or can be replaced by a high-value linearising resistor. It must not be populated alongside a real NTC — see §3.6.

3.4 Source capability detection

The default is USB500, so detection only ever has to answer one question: **is this a ≥ 1.5 A charger?** Everything else falls back safely.

CC sensing (C-to-C only). R36 / R37 tap CC1/CC2 through 1 kΩ into ADC1. With $R_d = 5.1 \text{ k}\Omega$:

Source Rp	V_{CC}	Advertised
56 kΩ	0.42 V	Default USB

Source Rp	V _{CC}	Advertised
22 kΩ	0.94 V	1.5 A
10 kΩ	1.69 V	3.0 A

Thresholds: $>0.2\text{ V}$ attached, $>0.66\text{ V} \geq 1.5\text{ A}$, $>1.23\text{ V} = 3.0\text{ A}$. Read **both** pins — only one carries the Rp, the other is open or is VCONN, and the pair also gives plug orientation. **No filter capacitor on the CC nodes:** the spec caps sink CC capacitance at $\sim 200\text{ pF}$. Multisample in software instead. Re-read periodically; a source may change its advertisement at any time.

The limitation that shapes the architecture: a USB-A-to-C cable is *required* by the spec to contain a $56\text{ k}\Omega$ Rp, so CC always reports Default USB no matter how large the charger behind it. For the expected majority case — an A-to-C cable — CC sensing tells you nothing.

BC1.2 (not fitted). The only mechanism that unlocks a USB-A wall charger, by detecting that D+ is shorted to D- through $\leq 200\text{ }\Omega$ (a Dedicated Charging Port). Needs a detector IC — BQ24392 or MAX14636 — whose integrated D+/D- isolation switch also solves the problem that the CH340K sits on that bus. Deferred; the EN1/EN2 GPIO control above is what keeps it a firmware-plus-one-IC change rather than a respin.

Guard rail. ILIM mode has no VIN-DPM, so a wrong detection has no graceful foldback. R38 / R39 ($100\text{ k} / 47\text{ k}$) divide VBUS to USB_VBUS_SENSE — $5.0\text{ V} \rightarrow 1.60\text{ V}$, $5.5\text{ V} \rightarrow 1.76\text{ V}$ — with C53 100 nF at the tap.

Firmware reverts to USB500 if VBUS sags below $\sim 4.5\text{ V}$. The divider only draws ($34\text{ }\mu\text{A}$) when USB is present, so it costs nothing on battery.

100 nF rather than the $1\text{ }\mu\text{F}$ used on BAT_ADC: this is a protection function with a deadline. Against the divider's $32\text{ k}\Omega$ Thévenin impedance it gives $\tau = 3.2\text{ ms}$, so a 100 ms poll sees a collapse $\sim 30\tau$ after it happens. A larger cap would average over a third of the reaction window.

Do not probe adaptively — stepping up to ILIM mode and watching what happens. Without VIN-DPM the failure mode is a brownout, not a graceful foldback.

Accepted risk — CC pins have no ESD protection. The USBLC6 covers D+/D- and VBUS only. CC1 and CC2 are exposed contacts in the receptacle and are directly touchable, so IEC 61000-4-2 contact discharge is a live threat. The $1\text{ k}\Omega$ series resistors (R36 / R37) protect the ADC pins but not R4 / R5 or the connector node itself. They also bound a VBUS-to-CC short — which the USB-C spec requires sinks to survive — to $(5 - 3.6)/1\text{ k} \approx 1.4\text{ mA}$ of clamp injection at the ESP32, which the pin tolerates; a faulty 20 V source would push 16 mA , which it does not. **Decision (2026-08-30): not mitigated in Rev B.** The fix, if it is ever wanted, is a low-capacitance TVS array on CC1/CC2 placed at the connector ahead of R36 / R37 — sub-pF against the $\sim 200\text{ pF}$ sink CC budget, and low leakage so it does not corrupt the resistive advertisement. TPD4E05U06 is the obvious candidate. Revisit if field units show USB-C detection failures.

3.5 Safety timers

R_TMR = $56\text{ k}\Omega$ gives a 7.5 h fast-charge timer (5.6 h worst case) and a 45 min pre-charge timer. Leaving TMR floating selects the internal default of $5\text{ h typ} / 4\text{ h min}$; grounding it disables the timers entirely.

The timer must outlast the slowest legitimate charge. Worst case is a 3000 mAh cell charging in USB500 mode while playing: $\sim 304\text{ mA}$ of charge current against 742 mA programmed is a 0.41 reduction, so the 5.6 h timer counts at $0.41\times$ and covers 13.7 h of wall clock against an 8.9 h charge.

On expiry the part latches a fault and /CHG flashes at $\sim 2\text{ Hz}$. Firmware should decode that — it is also how a TS fault surfaces, and it is otherwise invisible.

3.6 Battery temperature qualification

Mandatory in hardware, not firmware. `TS` sources 75 μ A into the pack thermistor and compares:

```
V_HOT = 300 mV → R_TS ≤ 4.0 kΩ → too hot (≈50 °C on a 10 k 103AT-2)
V_COLD = 2100 mV → R_TS ≥ 28 kΩ → too cold (≈ 0 °C)
```

The valid window is therefore **4.0 k Ω – 28 k Ω** , which a bare 10 k Type-2 NTC maps onto 0–50 °C with 3 °C hysteresis. Any added network shifts those trip points: a 10 k Ω resistor in parallel with the NTC moves the hot trip to ~33 °C, and a 1 k Ω resistor pins `R<sub>TS</sub>` below 1 k Ω permanently, inhibiting all charging. **Fit the pack NTC or `R_NTC_LIN`, never both.**

3.7 Status and state of charge

`/PGOOD` and `/CHG` are open-drain, pulled up to `3V3_A` (not `SYS`) so both nets are clean 3.3 V logic a GPIO can read directly — pulling them to the 4.4 V `SYS` rail would exceed the ESP32's input rating.

<code>/PGOOD</code>	<code>/CHG</code>	State
Hi-Z	Hi-Z	Running on battery
Low	Low	Charging
Low	Hi-Z	Charge complete, or suspended
Low	2 Hz flash	Fault — safety timer expired, or TS out of range

`R28` / `R29` are 330 Ω , giving ~2.7 mA per LED at 3.3 V — visible, and inside the 5 mA the `/PGOOD` and `/CHG` open-drain outputs are specified to sink. Both LEDs are dark on battery, since the pins go high-impedance with no input present.

State of charge is voltage-based: `R34` / `R35` form a 100 k / 100 k divider from `BAT` to ground, tapped as `BAT_ADC` (4.2 V → 2.1 V, inside ADC1's usable 150–2450 mV window at 11 dB attenuation). It costs 21 μ A continuously, ~15 mAh/month — negligible against the cell.

C52 100 nF at the tap is not optional, and it is not impedance matching. It does two jobs:

- **Charge reservoir.** The ESP32's ADC is successive-approximation: closing the sampling switch requires an internal sample-and-hold capacitor of a few pF to charge to the input voltage *through the external source impedance*, in a few microseconds. The divider's Thévenin impedance is 100 k || 100 k = **50 k Ω** , five times Espressif's <10 k Ω guideline, so without a local cap each conversion drags the node down and the converter digitises the droop. At ~10⁴ times the S/H capacitance, **C52** supplies that charge with no measurable sag and refills through the 50 k Ω between samples.

- **Low-pass filter.** $f_c = 1/(2\pi \times 50 \text{ k} \times 100 \text{ nF}) = \mathbf{31.8 \text{ Hz}}$, $\tau = 5 \text{ ms}$. `BAT_ADC` is a high-impedance trace crossing the board — exactly the node that collects switching and RF noise — and cell voltage moves over minutes, so heavy filtering is free accuracy. Allow ~25 ms after power-up before trusting a reading.

The alternative was a lower-impedance divider: 10 k / 10 k needs no cap but draws 210 μ A from the cell permanently instead of 21 μ A. The capacitor is the cheaper trade.

Calibrate with `esp_adc_cal` and the eFuse `Vref`, and multisample; raw ADC error is $\pm 6 \%$, which is $\pm 250 \text{ mV}$ of cell voltage and useless.

Two limits define what this can deliver. The reading is meaningless while charging — the charger is driving V_{BAT} toward 4.2 V — so firmware uses the `/PGOOD` + `/CHG` state instead and snaps to 100 % on termination. And the Li-ion OCV curve is flat from roughly 20 % to 80 % SoC, so voltage alone gives $\pm 10\text{--}15\%$ there. **This is a four-segment gauge and a dependable low-battery warning, not a trustworthy percentage.** If a real percentage is ever needed over AVRCP, add a MAX17048 to the existing I²C bus.

3.8 Power-related GPIO map

The budget is essentially full. Allocate deliberately.

Signal	Dir	Pin class
EN1, EN2	out	any GPIO (<code>GPI016</code> / <code>GPI017</code> ; avoid strapping pins)
AUDIO_VCC_EN	out	any GPIO. Pulled low by <code>R33</code> , so the codec rail is off at boot
CH340K gate	out	any GPIO — gate on a GPIO rather than <code>/PGOOD</code> , so firmware can isolate the bridge during BC1.2 detection
<code>/PGOOD</code> , <code>/CHG</code>	in	any input pin
BAT_ADC , USB_CC1 , USB_CC2 , VBUS sense	in	ADC1 only — ADC2 is unusable while the radio is active

Four output-capable pins remain free (`GPI04` , `16` , `17` , `21`) against exactly four outputs needed. Input-only ADC1 pins `GPI034` , `36` , `39` , plus `GPI032` / `33` , cover the inputs. There is no slack.

Safety. Charging is inhibited outside the cell's temperature window in hardware via the NTC on `TS` — not in firmware. Cell is removable; device is not stored in the vehicle.

Why not LFP: 3.2 V nominal sits below the 3.3 V rail for most of its discharge, forcing a boost converter in front of the codec.

4. Audio subsystem

ES8388 stereo codec — 24-bit, 96 dB DAC dynamic range, -83 dB THD+N. Chosen over the ESP32's internal DAC for quality, and over a DAC-only part because HFP needs an ADC.

Signal	Connection
Control	I ² C, address 0x20 (<code>CE</code> pulled to DGND)
Audio	I ² S, codec as slave ; full duplex for HFP
Clocks	MCLK, SCLK, LRCK from ESP32
Output	LOUT1/ROUT1 → 1 μ F → 10 Ω → analog switch → jack
Input (TX)	Jack → switch → LIN2/RIN2
Input (mic)	Differential LIN1/RIN1

All four codec rails run at 3.3 V. DVDD/PVDD at 1.8 V would put the codec's I/O in a different logic domain from the 3.3 V ESP32 — see §7.

Part	Function	Key spec
ES8388	Stereo codec	QFN-28, 0.4 mm; AVDD/HPVDD/DVDD/PVDD all 3V3

Part	Function	Key spec
SPH8878LR5H-1 (C3171733)	MEMS microphone	LGA-6, differential or single-ended, MSL 1
TS5A23159DGSR (C42751)	TX/RX path switch	Dual SPDT, 1 Ω R_{on} , break-before-make
3.5 mm TRS jack	Audio I/O	3-pole

Output coupling. 1 μ F blocks the VMID bias. Into a $\sim$ 10 k Ω AUX load that is a $\sim$ 16 Hz corner. Use a case size large enough that Class-II dielectric distortion stays below the codec's own THD.

Ground loop. Chassis ground (USB charger) and head-unit audio ground form a loop → alternator whine. Mitigations: run on battery with USB unplugged (free), or fit 1:1 isolation transformers in place of R7/R8. Both provisioned; footprints accept either.

5. Microcontroller & RF

ESP32-D0WD-V3. QFN-48, 0.35 mm pitch, dual-core Xtensa LX6.

Selected for firmware capability, not silicon features: it is the only Espressif part whose radio hardware supports **Bluetooth Classic**, and the only one whose SDK provides an **A2DP sink** and **HFP**. ESP32-S3/C3/C6/H2 and the entire Nordic nRF range are BLE-only. This choice follows directly from §1's compatibility requirement.

Part	Function	Key spec
ESP32-D0WD-V3	MCU + radio	BT Classic + BLE + Wi-Fi; needs external flash
W25Q32JV	Program store	4 MB, 3.3 V (JV not JW), on VDD_SDIO
40 MHz crystal	Reference	$\pm$ 10 ppm required; load caps per §7
PCB inverted-F antenna + C-L-C match	2.4 GHz	50 Ω controlled impedance, 4-layer

RF front end is a deliberate learning exercise. A pre-certified module would remove the matching network, crystal trim and impedance routing — that is precisely the work being kept. Requires a NanoVNA (antenna S11) and a frequency-offset measurement to close.

Stackup: 4-layer. Signal / GND / PWR / Signal, with a solid ground plane under the RF section.

6. Programming & debug

No USB peripheral on this ESP32 variant, and no SWD (Xtensa, not ARM).

Path	Parts	Use
USB auto-program	CH340K + UMH3N dual transistor on DTR/RTS	Day-to-day flashing
Off-board serial	J3 header (DNP)	Fallback
JTAG	J4 2×5 header (DNP), ESP-PROG	Debugging

Headers are DNP — fit before a debug session. 13 test points cover I²C, mic, audio out, boot, GPIO and rails.

7. Key calculations

Power budget

Load	Rail	Current
ESP32, BT Classic streaming (avg)	A	130 mA
ESP32 transmit peak	A	250 mA
SPI flash	A	5 mA
CH340K (USB attached only)	A	10 mA
2× LED @ 330 Ω	A	8 mA
Rail A		155 mA avg · 280 mA peak
ES8388 @ 3.3 V, playback + record	B	18 mA
MEMS mic	B	0.4 mA
Rail B		~20 mA

LDO A thermal. Bluetooth bursts last 1–2 ms; package time constant is seconds, so the junction integrates to the average, not the peak.

$$P_{\text{avg}} = (4.4 - 3.3) \times 0.155 = 0.17 \text{ W}$$

$$\text{SOT-89, } \theta_{JA} \approx 100 \text{ }^{\circ}\text{C/W} \rightarrow \Delta T = 17 \text{ }^{\circ}\text{C} \rightarrow T_j \approx 87 \text{ }^{\circ}\text{C at } 70 \text{ }^{\circ}\text{C ambient} \quad \checkmark$$

$$\text{SOT-23-5, } \theta_{JA} \approx 200 \text{ }^{\circ}\text{C/W} \rightarrow \Delta T = 34 \text{ }^{\circ}\text{C} \rightarrow T_j \approx 104 \text{ }^{\circ}\text{C} \quad \checkmark \text{ but tight}$$

LDO A dropout → usable battery. Dropout, not current, is the binding spec.

$$\text{Cutoff} = 3.3 \text{ V} + V_{\text{dropout}}(280 \text{ mA})$$

$$150 \text{ mV} \rightarrow 3.45 \text{ V} \rightarrow \sim 87 \% \text{ of cell}$$

$$250 \text{ mV} \rightarrow 3.55 \text{ V} \rightarrow \sim 83 \%$$

$$400 \text{ mV} \rightarrow 3.70 \text{ V} \rightarrow \sim 75 \%$$

$$1.2 \text{ V} \rightarrow 4.50 \text{ V} \rightarrow \text{will not run on battery} \quad (\text{NCP1117, Rev A - removed})$$

Runtime (LDO path, ~83 % usable): 1000 mAh $\approx$ 4.8 h · 2000 mAh $\approx$ 9.7 h · 3000 mAh $\approx$ 14.5 h.

Charge current and time. `R_ISET` programs 742 mA, but the input limit is what actually binds in every mode except ILIM. CC delivers ~85 % of capacity; the CV tail to a 75 mA termination adds ~0.5 h.

Mode	Input	System	Charge	2000 mAh	3000 mAh
USB500, idle	475 mA	~20 mA	455 mA	4.2 h	6.1 h
USB500, playing	475 mA	171 mA	304 mA	6.1 h	8.9 h
ILIM, either	1.073 A	20–171 mA	742 mA (ISET binds)	2.8 h	3.9 h

Charger thermal. Linear, so the charger burns $(V_{\text{IN}} - V_{\text{BAT}}) \times I_{\text{CHG}}$ plus the pass-FET loss. Worst case is the start of fast charge, $V_{\text{BAT}} \approx 3.0 \text{ V}$, in ILIM mode:

```

P = (5.0 - 4.4) × 0.913 + (4.4 - 3.0) × 0.742 = 0.55 + 1.04 = 1.59 W
VQFN-16 RGT, θJA = 44.5 °C/W → ΔT = 71 °C → Tj ≈ 96 °C at 25 °C ambient ✓
                                         Tj ≈ 116 °C at 45 °C ambient ✓ (reg. point 12)
USB500 mode: P = 0.71 W → ΔT = 32 °C → Tj ≈ 57 °C ✓

```

The input current limit is what guarantees this — raising `R_ISET` cannot overheat the part, because `ILIM` caps the current before it gets there.

Crystal load capacitance.

```

CL = (C1 × C2)/(C1 + C2) + C_stray      C_stray ≈ 2–5 pF

Rev A: 18 pF || 18 pF → 9 + 3 ≈ 12 pF against a 10 pF crystal → over-loaded, frequency low
Rev B: ≈ 15 pF || 15 pF → 7.5 + 3 ≈ 10.5 pF, then trim on measured offset (±10 ppm required)

```

Logic-level compatibility (why all codec rails are 3.3 V).

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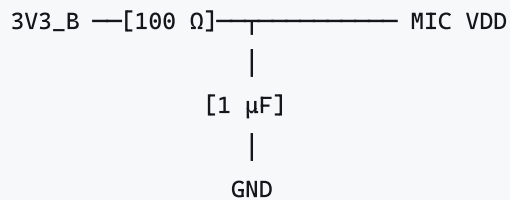
ESP32   VIH(min) = 0.75 × 3.3 = 2.475 V      VIL(max) = 0.825 V
Codec at PVDD 1.8 V: VOH ≈ 1.8 V → lands between VIL and VIH = undefined
Codec abs-max input = DVDD + 0.3 V = 2.1 V → 3.3 V drive exceeds it

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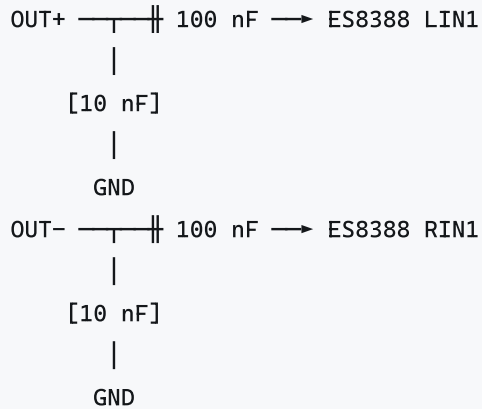
Audio output coupling. $f = 1/(2\pi \cdot 10 \text{ k}\Omega \cdot 1 \text{ }\mu\text{F}) \approx 16 \text{ Hz}$.

8. Conceptual circuits

Microphone — differential, no external bias. MEMS outputs are internally biased (0.66/0.70 V, 340/410 Ω). An electret-style bias resistor fights the internal amplifier and costs headroom.



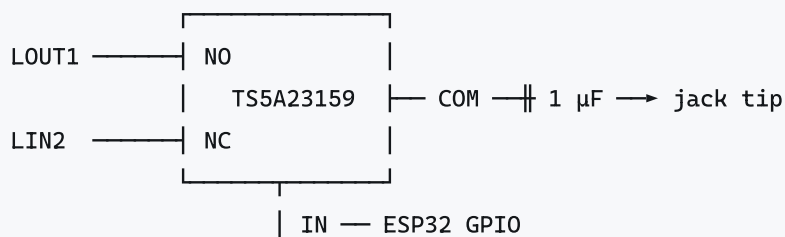
RC filter: mic PSRR is only -45 dB



Route OUT+/OUT- as a matched parallel pair - common-mode rejection depends on both picking up identical noise.

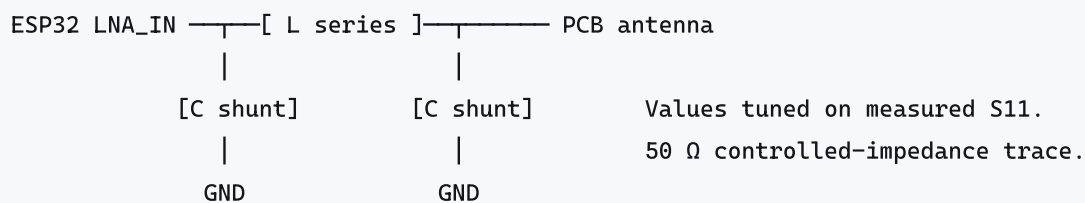
$10\text{ nF} + 340\text{ }\Omega \approx 47\text{ kHz}$ - RF shunt only, above the audio band.

TX/RX switch — placement is critical.



Switch sits on the **codec side** of the coupling capacitor. Both throws are at VMID (~1.65 V), inside the switch's 0–3.3 V supply rails. On the jack side the signal is centred at 0 V and every negative half-cycle would clip.

RF match — C-L-C Pi, per Espressif.



Values tuned on measured S11.
50 Ω controlled-impedance trace.

9. Firmware dependencies

Hardware choices that constrain firmware, or vice versa.

Item	Note
Bluetooth Classic	Only the original ESP32 has Classic radio hardware. A2DP sink + HFP come from ESP-IDF/ESP-ADF. Drives the entire MCU selection.
Pairing persistence	Bluedroid writes bonding keys to NVS in flash automatically. Survives power-off with no battery. Requires an initialised NVS partition.
MCLK pin	ESP32 can emit MCLK only on GPIO0/1/3. GPIO1/3 are the UART, so GPIO0 — which is also the BOOT strapping pin. Safe: the codec's MCLK input is high-Z at reset.
ADC	Use ADC1 only (battery sense). ADC2 is unusable while the radio is active. GPIO32–39 free.
Flash mode	Configure DIO . QIO requires IO2/IO3 correct — fixed in Rev B, but DIO is the safe default.
JTAG + GPIO12	MTDI is the VDD_SDIO strapping pin. A debugger driving TDI high across reset selects 1.8 V and the 3.3 V flash fails. Fix: burn <code>XPD_SDIO_*</code> eFuses to force 3.3 V. Irreversible .
Codec I²C address	0x20 (<code>CE</code> low). ESP-ADF's ES8388 driver default.
I²S full duplex	Required for HFP. Configure both TX and RX channels.
TX/RX switch	One GPIO selects the audio path. Firmware must track mode.
Deep sleep	If used for battery standby, populate the CAP2 RC network.
Charger limit sequencing	Boot default is USB500 (\$3.2). Firmware may raise to ILIM mode only after confirming a ≥ 1.5 A source. Drive <code>EN1</code> open-drain; leave both pins alone when <code>/PGOOD</code> is high-Z.
CC read	<code>USB_CC1</code> / <code>USB_CC2</code> on ADC1 . Read both, take the one in a valid band. An A-to-C cable always reports Default USB — do not treat that as a fault.
<code>/CHG</code> fault	A ~2 Hz flash is a latched fault (timer expiry or TS out of range), not "charging". Decode it; clear by toggling <code>CE</code> or the input.
SoC blindness	<code>BAT_ADC</code> is meaningless while charging. Report charger state instead, and snap to 100 % when <code>/CHG</code> releases.
Codec rail	<code>AUDIO_VCC_EN</code> is pulled low, so LDO B is off at reset . Firmware must enable it before touching the codec on I ² C/I ² S, or the ESP32 will back-power the ES8388 through its ESD diodes.
CH340K gate	Bridge is on a GPIO-controlled switch. Keep it off when USB is absent (~10 mA of battery budget) and during any BC1.2 detection.

10. Deltas from Rev A (as built)

#	Change	Reason
1	Codec rails 1.8 V → 3.3 V; delete 1.8 V LDO, add dedicated codec LDO	Logic-level incompatibility + supply noise
2	<code>CE</code> pulled to DGND	Was floating — indeterminate I ² C address
3	Add 10 nF on ESP32 <code>CAP1</code>	Required by datasheet; was absent
4	Swap flash IO2/IO3	Crossed; corrupts QIO reads
5	Delete mic bias network (R21/R22/R23/C43/C44); add RC on mic VDD	Electret circuit on a self-biased part
6	Mic → SPH8878LR5H-1	ICS-40720 discontinued

#	Change	Reason
7	LED resistors 1 k Ω → 330 Ω	~0.3 mA was invisible
8	TRRS → 3-pole TRS jack	4th pole unused
9	Add TX input path + analog switch	TX mode now in scope
10	Add battery charger, NTC, cell connector, VBAT sense	Battery in scope
11	NCP1117 → low-dropout LDO in SOT-89	1.2 V dropout cannot run from a cell
12	Crystal load caps 18 pF → ~15 pF, then trim	Over-loaded; ± 10 ppm required
13	Charger input limit made firmware-selectable (<code>EN1</code> / <code>EN2</code> on GPIOs, 100 k Ω pulls)	Fixed strap cannot adapt to source capability; default USB500 boots on a flat cell
14	<code>/PGOOD</code> , <code>/CHG</code> pull-ups moved SYS → 3V3_A and routed to GPIOs	4.4 V exceeds the ESP32 input rating; status was LED-only
15	Add <code>BAT_ADC</code> divider, CC1/CC2 sense taps	State of charge (REQ-PWR-10) and source detection
16	Gate the CH340K on a GPIO	~10 mA of battery budget with USB absent

Still open: cell choice (2000 mAh pouch vs 3000 mAh 18650) and therefore charge time; whether to fit BC1.2 detection; ground-loop mitigation (transformers vs battery-only playback); crystal offset measurement; RF match tuning; BOM line consolidation.

Accepted risks: CC pins carry no ESD protection (§3.4) — decided 2026-08-30, not mitigated in Rev B.

Not yet on the schematic (§3 describes the target): the

`EN1` / `EN2` / `/PGOOD` / `/CHG` / `BAT_ADC` / `USB_CC*` / `USB_VBUS_SENSE` global labels have no counterparts on the micro sheet yet; the CH340K load switch is not drawn; and `+3V3_SYS` is still a separate net from the micro sheet's `+3V3`. The power sheet itself is otherwise complete.

11. Manufacturing

JLPCB, 4-layer, **Economic PCBA** with the ESP32 excluded from assembly and hand-populated (it is the only Standard-Only part). All other parts verified Economic-tier.

Constraints: single-sided placement (satisfied — nothing on the back is assembled), 0402 minimum passive, 0.4 mm minimum IC pitch.

Build: 5 assembled + 5 bare. Target \$20 USD landed per unit.

Hand assembly: 0.35 mm-pitch QFN requires a laser-cut stencil, plugged/capped via-in-pad on the thermal pad (open vias wick solder and degrade the RF ground), and ~50–70 % paste coverage on the exposed pad.

12. References

Local (`hw/datasheets/`, `hw/app notes/`)

- ESP32 datasheet · ESP32 Hardware Design Guidelines · ESP32 Technical Reference Manual
- ES8388 datasheet rev 5.0 · ES8388 User Guide
- W25Q32JV datasheet rev G

- [ESP32-LyraT v4.3 schematic](#) — reference design for the codec section
- [AN043 2.4 GHz PCB Antenna](#) · Antennas for IoT

External

- [JLCPCB PCB assembly capabilities](#)
- [ESP-ADF ES8388 driver](#)
- [esptool](#) — set flash voltage eFuses

Project

- [0](#) — requirements, constraints, traceability
- [0](#) — netlist audit and findings
- [0](#) — battery architecture
- [0](#) — mic theory and selection

13. Revision history

Rev	Date	Change
A	2025-04-05	As-built schematic, commit <code>369c5f9</code>
B	2026-08-24	This document created. Scope decisions resolved; 12 deltas in §10
B	2026-08-30	§3 expanded to cover the full charger configuration: power-path loops, EN1/EN2 strapping, charge programming, source detection, safety timers, TS qualification, SoC and GPIO map. Deltas 13–16
B	2026-08-30	Power sheet complete: <code>R_ITERM</code> / <code>R_TMR</code> populated, LEDs to 330 Ω , ADC anti-droop caps and VBUS sense divider added. CC ESD recorded as an accepted risk